

I. XIII. COUPLING RATIO CONSISTENCY

The robustness of the entire mass spectrum calculation, including both stable and unstable states, relies on the success of the Zero-Parameter Economy, which fixed the dimensionless coupling ratio at $\mathbf{R} \approx \mathbf{0.007304}$ [SUPP 5]. This value must be maintained consistently across the $N \geq 1$ calculations to ensure the predicted masses are dimensionally consistent.

II. XIV. MODELING EXCITATION DYNAMICS AND DECAY PREDICTION

This Supplemental Material details the mechanism within the Scalar-Angular-Torsion (SAT) framework used to predict the dynamics and stability of both ground states ($\mathbf{N} = \mathbf{0}$) and dynamically unstable excited particle states ($\mathbf{N} \geq \mathbf{1}$). This predictive capability relies fundamentally on the String/Filament Dynamics component embedded within the **Total Covariant Action** (S_{SAT}).

A. A. THE QUADRATIC MASS SPECTRUM FORMULA

Excited particle states ($N \geq 1$) are governed by the same quadratic mass spectrum formula utilized for ground state stability checks:

$$m_{\text{string}}^2 = k \cdot (N + \beta \cdot Q^2 - a) \quad (1)$$

The predicted mass-squared values for the first excited oscillator level ($\mathbf{N} = \mathbf{1}$) for the stable topological charges ($\mathbf{Q} \leq \mathbf{3}$) are confirmed to be:

- $\mathbf{Q} = \mathbf{1}, \mathbf{N} = \mathbf{1} \rightarrow \mathbf{1.1}$
- $\mathbf{Q} = \mathbf{2}, \mathbf{N} = \mathbf{1} \rightarrow \mathbf{1.4}$
- $\mathbf{Q} = \mathbf{3}, \mathbf{N} = \mathbf{1} \rightarrow \mathbf{1.9}$

B. B. FILAMENT DISPLACEMENT AND DECAY CHANNELS

The geometry of the filament bundle dictates the likelihood and nature of particle decay. The decay channels and interaction cross-sections for unstable resonances ($\mathbf{N} \geq \mathbf{1}$) are determined by modeling the filament's **transverse displacement** ($\mathbf{X}$ and $\mathbf{Y}$). This displacement is calculated based on the filament's specific list of **String Modes**:

$$\mathbf{X}(z, t) = \sum_n \alpha_x \sin\left(\frac{n\pi z}{L}\right) \cos(n\omega_0 t + \phi_x) \quad (2)$$

$$\mathbf{Y}(z, t) = \sum_n \alpha_y \sin\left(\frac{n\pi z}{L}\right) \cos(n\omega_0 t + \phi_y) \quad (3)$$

Here, z represents the position along the filament, t is time, n is the oscillator mode number, L is the length, ω_0 is the fundamental frequency, and the coefficients (α) and phases (ϕ) define the dynamic geometry of the excitation.

This modeling task predicts the **dynamic geometry** of excited particles, which is the necessary precursor for predicting **decay channels** and **interaction cross-sections** for these resonances.

C. C. STRUCTURAL CONSTRAINT ON HADRONIC STABILITY

The consistency of this modeling approach with the Strong Interaction sector requires that the underlying topological structure enforces the $\mathbb{Z}_3$ fusion rule. This rule provides the geometric reason for the stability of three-constituent particles (baryons), reinforcing consistency with Quantum Chromodynamics (QCD).

D. D. CONTEXT: STABLE STATE FOUNDATION

This dynamic modeling complements the ultimate empirical validation targets for the **novel stable states** ($\mathbf{N} = \mathbf{0}$), which are fully defined at $\mathbf{Q} = \mathbf{2}$ ($\approx \mathbf{0.5073} \times 10^{-27}$ kg) and $\mathbf{Q} = \mathbf{3}$ ($\approx \mathbf{0.3403} \times 10^{-27}$ kg). The necessary particle identity (Spin/Flavor) for these ground states is confirmed to be derivable via **Holonomy Quantization**.

III. XII. THEORETICAL CLOSURE AND FORMAL COMPLETENESS FRAMEWORK

This Supplemental Material consolidates the foundational philosophical constraints and the criteria required for achieving theoretical completion within the Scalar-Angular Torsion (SAT) framework.

A. A. ZERO-PARAMETER ECONOMY AND AESTHETIC CLOSURE

The fundamental structural constraint on the entire model is the **Zero-Parameter Economy** (Normalization Principle), which mandates the axiomatic introduction of **zero numerical parameters** into the fundamental structure [Manuscript Draft, SUPP 5]. All fundamental dimensionless constants and physical ratios, such as the calibrated coupling ratio $\mathbf{R} \approx \mathbf{0.007304}$, must be derived solely from internal geometry and topology [Manuscript Draft, SUPP 5].

This commitment serves as the "ultimate honesty test" for the framework and is central to the claim of **aesthetic closure** (mathematical inevitability) [Manuscript Draft, SUPP 5].

B. B. HIERARCHY OF THEORETICAL COMPLETENESS

The development of the SAT framework is structured around achieving several defined levels of theoretical completeness:

1. 1. Level I: Foundational Rigor

This level is satisfied by demonstrating rigorous geometric definitions and computational verification [Variational Methods and Geometric Emergence in SAT Theory].

* **Mathematical Constraints:** The ****Total Co-variant Action (S_{SAT})**** must define a mathematically "Well-defined path integral." This requires the action to be σ -finite over the discrete twist ensemble (τ) and ****Fréchet-differentiable**** over the continuous field space (ξ) [Variational Methods and Geometric Emergence in SAT Theory, SUPP 11]. * **Formal Verification:** The internal algebraic consistency and stability are confirmed via ****Lean 4 formal certification**** [Lean 4 Verified Mass Spec..., SUPP 5]. This process validates criteria such as ****Positive Effective Mass**** and the ****Topological Dominance Condition**** ($\mathbf{R} \ll 1$) [SUPP 5].

2. 2. Level III: Informational Closure

This level requires providing an "informational story" to bridge classical and quantum domains [SUPP 3]. This is addressed by defining the ****Entropy Functional**** (S_{filament}) for the discrete twist ensemble (τ) [SUPP 3]. Furthermore, the model attributes entanglement and wave-function collapse to ****worldline backbleed**** [SUPP 3].

3. 3. Level IV: Semantic Unity

This represents the metaphysical closure gesture, achieved by formally verifying that the physics is reconstructible within a proof-assistant logic (Lean 4) [Manuscript Draft, Variational Methods and Geometric Emergence in SAT Theory].

C. C. MANUSCRIPT SUBMISSION AND DATA RIGOR

Given the reliance on formal proofs and calibrated numerical anchors, adherence to data standards is essential for publication in journals like ***Physical Review D*** (PRD).

1. **Data Availability Statement (DAS):** For manuscripts submitted to PRD on or after September

4, 2024, a DAS is required. This statement, automatically added after the Acknowledgments, must address the availability of the Lean 4 code and calibration data necessary to reproduce the predicted mass spectrum. 2. **Supplemental Material Reference:** References to the formal proofs and technical details provided across these Supplemental Materials must follow guidelines for linking to supplemental data, including DOIs or URLs upon final publication. 3. **Citation of Computational Tools:** References to software or research reports, such as the Lean 4 environment utilized for formal verification, should include DOIs or URLs. Astrophysics Source Code Library (ASCL) entries, if applicable to the cosmological models suggested by the predicted dark matter candidates, would follow a specific citation format.

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